

1. (1) Newton's iteration algorithm: $g(x) = x - \frac{f(x)}{f'(x)}$, $x_{n+1} = g(x_n)$

For the given $f(x)$, $g(x) = \frac{1 - 4(e^x - 1)\cos x + 2x^2 + \cos 2x}{4(e^x - 1)\sin x - 4e^x \cos x + 4x - 2\sin 2x}$, where $x_0 = 0.6$

$$\therefore x_1 = g(x_0) = 0.425617, \quad x_2 = g(x_1) = 0.445287$$

$$x_3 = g(x_2) = 0.445415, \quad x_4 = g(x_3) = 0.445415$$

2. (1) the Lagrange interpolation polynomial basis terms of $f(x)$ is:

$$l_i(x) = \prod_{0 \leq j < n, j \neq i} \frac{x - x_j}{x_i - x_j}$$

$$l_0(x) = \frac{x-0}{-1-0} \cdot \frac{x-3}{-1-3} = \frac{x(x-3)}{4}$$

$$l_1(x) = \frac{x+1}{0+1} \cdot \frac{x-3}{0-3} = -\frac{(x+1)(x-3)}{3}$$

$$l_2(x) = \frac{x-0}{3-0} \cdot \frac{x+1}{3+1} = \frac{x(x+1)}{12}$$

$$\left. \begin{array}{l} l_0(x) = \frac{x(x-3)}{4} \\ l_1(x) = -\frac{(x+1)(x-3)}{3} \\ l_2(x) = \frac{x(x+1)}{12} \end{array} \right\} L_2(x) = \sum_{i=0}^2 f(x_i) l_i(x) = 3 \cdot \frac{x(x-3)}{4} - 8 \cdot \frac{(x+1)(x-3)}{3} + 9 \cdot \frac{x(x+1)}{12}$$

$$(2) \text{ Approx } f(2) \text{ is } L_2(2) = 3 \cdot \frac{-2}{4} - 8 \cdot \frac{-3}{3} + 9 \cdot \frac{6}{12} = 11$$

$$3. (1) \text{ Normal Equation: } \begin{cases} n \cdot b + (\sum x_i) \cdot a = \sum y_i \\ (\sum x_i) b + (\sum x_i^2) a = \sum x_i y_i \end{cases} \Rightarrow \begin{cases} a = (n \sum x_i y_i - \sum x_i \cdot \sum y_i) / [n \sum x_i^2 - (\sum x_i)^2] \\ b = \bar{y} - a \bar{x} \end{cases}$$

$$\therefore n = 5$$

$$\sum x_i = 3.4212 + 3.9279 + 5.1779 + 6.8251 + 11.8603 = 31.2124, \quad \bar{x} = 6.24248$$

$$\sum y_i = 13.5339 + 15.1001 + 18.9670 + 23.6083 + 36.9070 = 108.1163, \quad \bar{y} = 21.6233$$

$$\sum x_i^2 = \dots \approx 241.1924$$

$$\sum x_i y_i = \dots \approx 802.6802$$

$$\therefore a = (5 \times 802.6802 - 31.2124 \times 108.1163) / (5 \times 241.1924 - 31.2124^2) = 2.7566$$

$$b = \bar{y} - a \bar{x} = \frac{1}{5} \times 108.1163 - \frac{1}{5} \times 31.2124 \times 2.7566 = 4.4152$$

$$E = \|Ax - b\| = \sqrt{\sum (ax_i + b - y_i)^2} \approx 0.6164$$

$$4. P(x) = -0.0029x^2 + 0.5289x + 1.2914$$